

Omri Nisan Solan - CV

I am a Bourgain Fellow and an associate research scholar at Princeton University and at the Institute for Advanced Study.

Employment:

- 2025 - 2029: Bourgain Fellow and an Associate Research Scholar at Princeton University and at the Institute for Advanced Study.

Education:

- 2022 - 2025: Ph.D. student at the Hebrew University of Jerusalem, under the supervision of Prof. Elon Lindenstrauss.
- 2015 - 2021: Master's student at Tel Aviv University, under the supervision of Prof. Barak Weiss (Summa cum laude).
- 2010 - 2015: Bachelor of Science, Mathematics, Tel Aviv University (Summa cum laude).

Research Interest:

My primary research focus is on homogeneous dynamics. I leverage tools from number theory, probability, and algebraic topology to address problems in this field.

Teaching:

- 2026 - Instructor, Princeton university, "Multivariate calculus" (MAT 201).
- 2024 - Teaching Assistant, Hebrew University of Jerusalem, "Fundamental concepts in commutative algebra and algebraic geometry".
- 2023 - Teaching Assistant, Hebrew University of Jerusalem, "Fundamental concepts in spectral analysis".

Awards:

- Nessyahu Prize in Mathematics, for excellent Ph.D. dissertations in the sciences (2026), awarded by the Israel Mathematical Union.
- Aharon Katzir Prize for excellence in Ph.D. studies in experimental sciences at the Hebrew University (2026).
- Lior Tzafriri Award for excellence in Ph.D. studies (2025).
- Award of excellence for Master's Studies at Tel Aviv University (2021).
- International Mathematics Competition for University Students, 2014 First Prize, 2015 Grand First Prize & Prize for special solution, 2017 First Prize.
- Dean's award for distinguished students - Tel Aviv University (2015).
- International Mathematical Olympiad, 2012 silver medal, 2013 gold medal, 2014 silver medal.

Active Participation in Scientific Meetings:

- 2026 - “From billiard dynamics to infinite volume dynamics”, Nessyahu Prize Talk - Annual Meeting of the Israel Mathematical Union.
- 2026 - “Periodic billiards and invariant measures on translation surfaces” - ‘Group Actions and Harmonic Analysis in Number Theory’ workshop, Oberwolfach, Germany.
- 2026 - “Horocycle-invariant measures on the moduli space of translation surfaces” - ‘Homogeneous Dynamics and Anosov Representations’ workshop, SLMATH, Berkeley.
- 2026 - “Horocycle-invariant measures on the moduli space of translation surfaces” - University of Houston Dynamical Systems Seminar.
- 2026 - “Horocycle-invariant measures on the moduli space of translation surfaces” - University of Chicago Dynamics Seminar.
- 2026 - “Equidistribution of expanding horospheres in the space of translation surfaces” - Joint IAS/PU Groups and Dynamics Seminar.
- 2025 - “Random walk on homogeneous spaces”, Groups & Dynamics - Hebrew University of Jerusalem, Israel.
- 2025 - “Random walks on the space of lattices”, Mathematics Colloquium - Tel Aviv University, Israel.
- 2025 - “Random walk on homogeneous spaces”, Midrasha on Groups - Weizmann Institute of Science, Israel.
- 2025 - “Gap in critical exponents”, ‘The Ergodic Method in Group Theory’ conference - Northwestern University.
- 2025 - “Critical exponent gap in hyperbolic geometry”, UC San Diego Group Actions Seminar.
- 2025 - “Returns in flat surface dynamics”, Tzafriri Talk - Hebrew University of Jerusalem, Israel.
- 2024 - “Limits of compact diagonal orbits in the lattice space”, ‘Dynamics and Number Theory’ workshop - Hebrew University of Jerusalem, Israel.
- 2024 - “Gap in critical exponents”, Joint IAS/PU Groups and Dynamics Seminar.
- 2024 - “Gap in critical exponents”, Group Actions, Geometry and Dynamics - Yale University.
- 2024 - “Gap in critical exponents”, Dynamical Systems Seminar - Penn State University.
- 2024 - “Gap in critical exponents”, ‘Recent Trends in Homogeneous and Teichmüller Dynamics’ workshop - Maryland.
- 2024 - “Gap in critical exponents of $SL_2(\mathbb{R})$ orbits in nonarithmetic quotients of $SL_2(\mathbb{C})$ ”, ‘Distribution of orbits: Arithmetics and Dynamics’ workshop - Paxmontana, Switzerland.
- 2024 - “Gap in critical exponents of $SL_2(\mathbb{R})$ orbits in nonarithmetic quotients of $SL_2(\mathbb{C})$ ”, ‘Group Actions with Hyperbolicity and Measure Rigidity’ workshop - Institut Henri Poincaré, Paris, France.
- 2024 - “Divergence of large dimensional diagonal orbits”, ‘Diagonal actions in the space of lattices’ workshop - École Polytechnique, Palaiseau, France.
- 2023 - “Limits of Compact Diagonal Orbits in the Lattice Space”, ‘Diophantine Analysis, Dynamics and Related Topics’ conference - Technion, Israel.
- 2022 - “Algebraic Divergence and Geometry of Symmetric Spaces”, Groups & Dynamics Seminar - Tel Aviv University, Israel.

- 2021 - “Geometry of the Cusp and Divergent Trajectories”, Dynamics & Probability - Hebrew University of Jerusalem, Israel.
- 2021 - “Parametric Geometry of Numbers with General Flows”, Diophantine Analysis and Related Topics, online conference.
- 2017 - “Stable and Well-Rounded Lattices in Diagonal Orbits” - Action Now Wandering Seminar, Ben-Gurion University, Israel.
- 2015 - “Divergent Trajectories in $SL_3(\mathbb{R})/SL_3(\mathbb{Z})$ ”, Dynamics & Probability - Hebrew University of Jerusalem, Israel.

Service to the Community:

- Referee reports for GAFA, Journal of the European Mathematical Society, Advances in Mathematics, and Mathematische Zeitschrift.
- 2025-2026 - Coorganizer of the Joint IAS/PU Groups and Dynamics Seminar.
- 2019-2021 - Tutoring member of the Israeli team to the International Mathematical Olympiad.
- 2015-2017 - Team in the Israeli Mathematical Olympiad Program.